PRODUCT REQUIREMENTS DOCUMENT (PRD) — VERSION 1.0

Tandem Discharge

Autonomous Inpatient & Urgent Care Clinical Synthesis Engine

Transforming multi-day fragmented ward round notes, laboratory feeds, and medication charts into NHS-compliant Electronic Discharge Notifications (eDN) in seconds.

Project:	Tandem Discharge (Clinical Admin Automation)	**Target Event:**	NXGN x Tandem Health Hackathon 2026
Author / Lead:	Engineering & Clinical Product Team	**Date:**	September 5, 2026
Target Audience:	NHS Junior Doctors, Ward Consultants, GPs, Urgent Care Leads	**Clinical Standards:**	PRSB eDN, NICE Guidelines, SNOMED-CT
Status:	READY FOR IMPLEMENTATION	**Target Execution:**	3-Hour Hackathon Build + Live Demo

Executive Summary & Strategic Thesis:
Tandem Health has pioneered ambient clinical scribing for single-encounter consultations. However, the single largest administrative bottleneck causing NHS hospital bed-blocking is the **Electronic Discharge Notification (eDN)** in secondary and urgent care. Junior doctors spend 30 to 60 minutes per patient manually synthesizing fragmented ward notes, blood test trends, and drug charts. **Tandem Discharge** expands Tandem's intelligence from ambient audio to multi-source asynchronous clinical synthesis—generating a fully reconciled, audit-traceable discharge package in under 90 seconds.

1. Problem Statement & Clinical Landscape

In NHS hospital trusts and urgent care centres across the United Kingdom, clinical documentation is acutely bifurcated between *consultation recording* and *episode synthesis*. While ambient AI scribes excel in primary care 1-on-1 outpatient appointments, inpatient care operates asynchronously across multiple days, multidisciplinary teams, and fragmented health IT silos.

1.1 The Inpatient Discharge Summary Crisis

Writing the Electronic Discharge Notification (eDN) is universally cited by NHS Junior Doctors (Foundation Year 1 & 2) as their most dreaded administrative task. When a patient is deemed medically fit for discharge (MFFD), the following manual workflow occurs:

- **Fragmented Record Scouring:** The clinician must read 3 to 10 days of physical and electronic notes, nursing entries, lab result trends, and microbiology cultures.
- **High-Risk Medication Reconciliation:** The clinician must cross-reference medications on admission against the inpatient drug chart (Kardex or ePMA) and formulate the discharge TTO (To-Take-Out) prescription. Up to 50% of medication errors in the NHS occur during clinical handovers.
- **Manual GP Instruction Formulation:** The clinician must draft clear, actionable follow-up items for primary care (e.g., repeat U&Es;, titration dates, outpatient clinic referrals).
- **Bed-Blocking & Flow Stagnation:** The discharge summary and pharmacy dispensing take 4 to 8 hours to complete. Beds cannot be turned over for acute admissions from A&E;, worsening ambulance handover delays.

Metric / Workflow Dimension	Current NHS Manual Process	With Tandem Discharge	Net Impact
Time per Discharge Summary	35 - 60 minutes	**< 90 seconds** (automated draft)	**95% time reduction**
Medication Reconciliation Accuracy	High risk (omissions in 30-40% of cases)	**Zero-omission diff matrix** with reasons	**Eliminates TTO errors**
Time to Free Hospital Bed	4 - 6 hours post-ward round	**Immediate** (ready by 10:30 AM)	**Unlocks hospital capacity**
GP Clarity & Safety Netting	Dense, unformatted narrative	**Structured, prioritized checklist**	**Zero lost follow-ups**
Patient Comprehension	Complex medical jargon	**Plain-English leaflet (Reading Age 11)**	**Reduces readmissions**

1.2 Stakeholder Ecosystem & Beneficiaries

- **Junior Doctors (FY1/FY2/ST):** Reclaims 2+ hours per shift previously spent typing notes; mitigates burnout.
- **Ward Consultants & Clinical Leads:** Instant visibility into patient trajectory with audit-verifiable source citations.
- **General Practitioners (Primary Care):** Receives standardized PRSB-compliant letters with highlighted actions rather than buried free-text.
- **Urgent Care & Hospital Operations:** Accelerates morning bed-turnover, directly addressing NHS winter pressures and four-hour emergency targets.

2. Strategic Alignment with Tandem Health

Tandem Health has established market leadership in ambient consultation recording. The strategic imperative for healthcare AI in 2026 is moving from *point-in-time scribing* to *episodic clinical intelligence*. **Tandem Discharge** represents the exact bridge from primary care scribing into secondary care and urgent care enterprise adoption.

2.1 Why Ambient Scribing Alone Cannot Solve Inpatient Discharge

Ambient listening only hears what is spoken aloud during a single bed visit. Inpatient care is asynchronous: a blood test is ordered on Monday, an ultrasound is performed on Tuesday, microbiology calls on Wednesday with blood culture results, and a medication is switched on Thursday. Scribing a 2-minute bedside conversation misses 90% of the relevant clinical context. **Tandem Discharge ingests the chronological timeline** of ward notes, observations, and charts to create a coherent medical narrative.

The Two Pillars of Tandem's Clinical Documentation Ecosystem:

1. Tandem Ambient (Existing Core): Synchronous audio capture during live 1-on-1 consultations. Optimized for GP clinics and outpatient specialty appointments.

2. Tandem Discharge (This Product): Asynchronous timeline synthesis across multi-day inpatient stays, lab feeds, and drug charts. Optimized for Acute Wards, Urgent Care Units, and Hospital Discharge Lounges.

2.2 Alignment with NHS & UK Standards

- **PRSB Compliance:** Formatted according to the Professional Record Standards Body (PRSB) Core Information Standard for eDN.
- **SNOMED-CT Diagnostic Encoding:** Automatically attaches validated clinical terms and ICD-10 codes for trust billing and QOF audit.
- **Caldicott & NHS Information Governance:** Zero training on protected health information; fully anonymized synthetic data for demonstration.
- **EHR Compatibility:** Formatted outputs engineered for seamless paste into EMIS Web, SystmOne, Epic Hyperspace, and Cerner Millennium.

3. Functional Requirements & Feature Scope

The system is architected around a dual-pane workspace: raw chronological inputs on the left, and multi-dimensional, synthesized clinical outputs on the right.

3.1 Feature Matrix (MoSCoW Prioritization for Hackathon)

Feature Module	Priority	Description & Clinical Rationale	Verification Criteria
Multi-Source Intake Feed	**P0 (Must)**	Chronological ingest of admission notes, ward round logs, nurse vitals, and lab feeds.	Accepts pasted text, pre-loaded cases, or raw JSON feeds.
AI Synthesis Engine	**P0 (Must)**	Generates structured NHS eDN: Chief Complaint, Hospital Course, Diagnoses, Procedures.	Outputs complete narrative in < 5 seconds with zero hallucination.
Interactive Med Rec Matrix	**P0 (Must)**	Visual diff table categorizing meds into Started, Stopped, Dose Changed, and Unchanged.	Mandatory 'Reason for Change' column for clinical safety.
Actionable GP Checklist	**P0 (Must)**	Structured primary care tasks with explicit timelines (e.g., 'Repeat U&Es; in 7 days').	Categorized by Urgency (Urgent, Routine, Safety Net).
Patient 'Take-Home' Leaflet	**P0 (Must)**	Translates medical jargon into plain English (Reading Age 11) with warning signs.	Includes daily visual medicine schedule and 111/999 rules.
Source-Tracing Tooltips	**P0 (Must)**	Clicking any synthesized sentence highlights the source line in the input timeline.	Provides 100% auditability for clinician verification.
EHR Copy & Export Modal	**P1 (Should)**	One-click formatted copy tailored for EMIS Web, SystmOne, and Epic Hyperspace.	Copies plain formatted text to OS clipboard instantly.
SNOMED-CT Auto-Coding	**P1 (Should)**	Attaches standardized SNOMED concept IDs to all extracted diagnoses.	Badges display concept codes alongside diagnostic text.

3.2 Deep-Dive: The Medication Reconciliation Matrix

Medication reconciliation is the single most safety-critical component of patient discharge. The system enforces a 4-state state machine for every drug identified across the patient admission:

1. **STARTED:** Newly initiated during admission (e.g., Co-Amoxiclav). Requires planned duration and indication.
2. **STOPPED:** Discontinued inpatient medication (e.g., Codeine ceased due to constipation). Requires explicit clinical rationale so GP does not restart.
3. **DOSE CHANGED:** Altered dosage or frequency (e.g., Furosemide doubled to 80mg OD). Requires titration monitoring instructions.
4. **CONTINUED:** Unchanged home medications maintained throughout stay (e.g., Atorvastatin 20mg ON).

4. Target User Personas & Clinical Workflow

The user experience is designed around the high-pressure environment of NHS acute wards and urgent care units.

Persona A: Inpatient Junior Doctor	Persona B: Urgent Care Clinical Lead	Persona C: Receiving GP
Dr. Alex Smith, FY1 *Ward 4B (Acute Medicine)* **Pain Point:** Has 6 patients ready for discharge by 11 AM. Writing summaries manually takes until 3 PM. Trapped behind computer while ward tasks pile up. **Needs:** Rapid synthesis of messy notes, automated drug diffs, instant copy to hospital EHR.	**Dr. Jonathan Tai** *Clinical Lead, NHS Urgent Care* **Pain Point:** High patient turnover. Urgent care handovers require rapid risk stratification, clear red flag safety nets, and seamless primary care follow-up. **Needs:** Standardized discharge templates, bulleted safety instructions, zero missing lab alerts.	**Dr. Sarah Patel, GP Partner** *Millwood Medical Practice* **Pain Point:** Receives 30 hospital discharge letters daily. Spends 10 minutes per letter hunting for what actually changed and what bloods she needs to order. **Needs:** Prominent 'GP Action Required' box on page 1 with specific deadlines.

4.2 End-to-End Clinical User Journey

1. **Morning Ward Round (09:00):** Consultant declares patient Medically Fit for Discharge (MFFD).
2. **One-Click Case Intake (09:15):** Junior doctor selects patient or pastes multi-day ward notes and Kardex snapshot into Tandem Discharge.
3. **Instant Synthesis (09:16):** AI engine generates PRSB eDN, extracts diagnoses, maps SNOMED codes, and performs 4-way medication reconciliation.
4. **Clinician Review & Verification (09:18):** Doctor inspects red flags, reviews med diffs, verifies GP action dates, and modifies text if needed.
5. **One-Click EHR Dispatch (09:20):** Summary is copied into hospital EHR, sent electronically to GP, and a patient leaflet is printed.
6. **Bed Turnaround (09:30):** Patient moves to discharge lounge 5 hours earlier than manual baseline, freeing bed for acute admission.

5. Clinical Safety, Governance & Guardrails

Clinical generative AI must operate under an uncompromising standard of safety. The system implements a four-tier guardrail architecture:

5.1 Guardrail Architecture

- **Human-in-the-Loop Verification:** The AI acts strictly as an assistant. Every section of the eDN is editable by the clinician. The document cannot be marked 'Dispatched' without explicit clinician review confirmation.
- **Strict Extractive Grounding:** The synthesis prompt enforces that zero diagnostic or therapeutic assertions are made unless explicitly present in the input chronological record. Speculation is disallowed.
- **Interactive Source Tracing:** Clicking on any synthesized paragraph highlights the exact sentence in the source ward notes, providing immediate visual verification.
- **Contradiction Detection:** If admission notes state 'NKDA' (No Known Drug Allergies) but an inpatient note mentions a Penicillin rash, the system generates a high-priority amber warning flag for clinician resolution.

Clinical Governance Statement (NHS DCB0129 / DCB0160 Alignment):
All test data utilized for the hackathon demonstration is 100% synthetic, clinically realistic, and strictly de-identified in compliance with the UK Data Protection Act 2018 and Caldicott Principles. No live NHS patient data is processed or transmitted.

5.2 Evaluation Metrics for Clinical Accuracy

- **Omission Rate:** 0% tolerance for missing discontinued or newly initiated medications.
- **Hallucination Rate:** 0% tolerated for non-documented clinical interventions.
- **Readability Index:** Patient Leaflet constrained to Flesch-Kincaid Grade Level ≤ 6 (Reading Age 11).

6. Hackathon 3-Hour Execution Roadmap & Pitch

With building beginning at 12:30 and the submission deadline at 15:45, time allocation is strictly phased:

Time Window	Phase	Key Deliverables & Action Items	Exit Criteria
12:30 - 13:15 (45 mins)	**Phase 1: Shell & Data Ingestion**	• Scaffold React/Tailwind shell with Lovable. • Embed 3 rich NHS mock patient cases (Frailty, Post-Op, Asthma). • Build left-hand chronological timeline feed.	Working split-pane UI with interactive patient selector.
13:15 - 14:15 (60 mins)	**Phase 2: Synthesis & Med Rec Matrix**	• Implement eDN generation state machine. • Build interactive 4-status Medication Reconciliation diff table. • Build structured GP Action checklist with urgency tags.	Working synthesis with dynamic tab switching.
14:15 - 14:45 (30 mins)	**Phase 3: Working Lunch & Intelligence**	• Connect LLM API / fine-tune zero-hallucination prompt. • Implement source-citation highlight tooltips. • Build patient 'Take-Home' leaflet view.	Live synthesis generating both clinical & patient views.
14:45 - 15:30 (45 mins)	**Phase 4: Design Polish & Export**	• Apply Tandem Health teal design system & micro-animations. • Implement EHR one-click copy modal (SystmOne/EMIS/Epic). • Add confetti celebration & demo reset toggle.	Pixel-perfect, wow-factor user interface.
15:30 - 15:45 (15 mins)	**Phase 5: Final Submission & QA**	• Deploy live URL to Vercel/Lovable. • Push code to public GitHub repo. • Complete official hackathon submission form.	**Submitted before 15:45 deadline**

The 2-Minute Live Pitch Strategy (For Team Demos at 15:55):
0:00 - 0:30 (The Hook): Introduce Dr. Alex Smith on Ward 4B at 3 PM, drowning under 6 discharge summaries while ambulances wait outside. State the thesis: *'Tandem Health solved the single consult. We built the engine that solves the hospital episode.'*
0:30 - 1:15 (The Live Demo): Click into an 82yo Heart Failure case with 4 days of chaotic ward notes. Hit *'Synthesize Discharge Package'*. Show the generated PRSB summary, the instant color-coded Med Rec diff table (Ramipril stopped, Furosemide doubled), and click a sentence to show instant source-tracing back to Day 2 ward round.
1:15 - 1:45 (The Magic Twist): Toggle to the **'Patient Leaflet'** tab. Show the warm, plain-English summary and daily medicine chart. Then click **'Copy to EHR'** (formatted for EMIS/SystmOne).
1:45 - 2:00 (The Close): Connect directly to Tandem's mission: freeing clinicians from admin across the entire patient journey. Close with an invitation for Q&A.;